Xun Zhang

Date of Birth: 1991.01 **Place of Origin:** China
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Education Background

Associate Researcher **Mar.2024-Present**
School of Atmospheric Science, Nanjing University
Research Focus: Global greenhouse gas inversion and uncertainty assessment

Ph.D. in Geography **Sep. 2017 – Mar. 2024**
International Institute for Earth System Science, Nanjing University
Ph.D. Thesis: Impact of Vegetation Fire on Spatial and Temporal Variations of Carbon Monoxide in Asia

Master in Ecology (Ecohydrology) **Sep. 2014 – Jun. 2017**
Center for Ecological Research, Northeast Forestry University
Master Thesis: Redistribution of nitrogen deposition and its nitrogen budget in temperate forest in the Maoershan region, Northeast China

Bachelor in Forestry (Forest Fire Prevention) **Sep. 2010 – Jun. 2014**
School of Forestry, Northeast Forestry University

Research Interests

Atmospheric physics and environment; Remote sensing of environment resources; Atmospheric chemistry and earth system science; Atmospheric chemical transport models; Greenhouse Gas Inversion

Research Experience

Mar. 2024 – Mar. 2026
School of Atmospheric Sciences, Nanjing University
Collaborating Advisor: Prof. Huilin Chen
China National Key R&D Program: Research on Methane Emission Assessment Methods and Standards Based on Satellite and Ground Monitoring

By employing two independent inversion systems (GEOS-Chem + analytical method, TM5 + 4DVAR) with identical configurations (observation data, prior inventories) to conduct inversions, a comparative analysis of the results is performed to characterize the uncertainties in methane (CH_4) emission flux estimates attributed to transport models and optimization algorithms across different spatiotemporal scales (global and key regions).

Based on the same prior methane emission fluxes, forward simulations of methane concentrations are conducted using GEOS-Chem and TM5. The vertical profiles and spatial distributions of tropospheric/stratospheric methane concentrations over global and key regions are compared. Sensitivity experiments (e.g., process switch-off) are designed to separately quantify the contributions of transport processes in GEOS-Chem and TM5: advection, convection, and diffusion, to CH_4 concentration transport.

Using GOSAT satellite data and NOAA global surface methane observations, global methane emission inversions for 2017 are performed with TM5-4DVAR and the GEOS-Chem analytical method. The spatiotemporal differences and uncertainties in CH_4 concentration and flux inversions due to variations in advection, convection, and diffusion between the two models are analyzed. Under the current observational data and model inversion framework, an evaluation scheme is proposed to more effectively calibrate and validate the CH_4 inversion errors of transport models.

Sep. 2017 – Mar. 2024
International Institute for Earth System Science, Nanjing University
Super Advisor: Prof. Jingxian Liu
China National Key R&D Program: Global High-Resolution Carbon Assimilation System Based on Multi-Source Satellite Remote Sensing

Carbon monoxide (CO) is a major air pollutant, the precursor of O_3 affecting the oxidation of atmospheric chemistry and the ozone dynamics, is a useful tracer for tracking global pollutant emissions and transport resulting from biomass burning. Asia is characterized by the highest CO concentrations in the world; however, the specific contribution of vegetation fire emissions to CO levels in this region has yet to be fully quantified. Using CO total column from the Measurement of Pollution in the Troposphere (MOPITT) and the Atmospheric Infrared Sounder (AIRS) on NASA's Aqua satellite, along with Moderate Resolution Imaging Spectroradiometer (MODIS) fire counts and Global Fire Emissions Database (GFED) CO emissions data, complemented by land cover data from the European Space Agency Climate Change Initiative and MODIS, this study elucidates vegetation fire emissions affecting the interannual

variability and trends of Asian CO total column from 2003 to 2017. Using a "tagged CO simulation" with a global chemical transport model, GEOS-Chem, this study assesses the contributions of vegetation fire-sourced CO (CO_{BB}) from various global regions to Asia's atmospheric CO levels between 2002 and 2020. Additionally, this research incorporates reanalysis data from the National Centers for Environmental Prediction (NCEP) and the National Center for Atmospheric Research (NCAR) to explore the El Niño-Southern Oscillation (ENSO) effect on the interannual variability of CO concentrations in Asia.

Utilizing ground-based observational data from monitoring stations (2013–2020), along with MOPITT and AIRS CO volume mixing ratio (VMR) and total column CO data, combined with HYSPLIT trajectory analysis, we analyze the impacts of biomass burning on surface $PM_{2.5}$, CO, NO_2, O_3 concentrations, as well as the ratios $\Delta PM_{2.5}/\Delta CO$, $\Delta NO_2/\Delta CO$, and $\Delta O_3/\Delta CO$ in Northeast China. The study further discusses the correlations among CO, NO_2, and O_3 and their influencing factors during the transport of air masses affected by biomass burning. Finally, we assess the health effects of air pollution caused by biomass burning on urban clusters in Northeast China.

Sep. 2014 – Jun. 2017

Center for Ecological Research, Northeast Forestry University

Super Advisor: Prof. Chuankuan Wang

Projects: CAS Field Station Alliance Project, MOE Changjiang Scholars and Innovation Team Development Plan

Conducted systematic studies on nitrogen fluxes (total nitrogen, soluble nitrogen, particulate nitrogen) in a temperate deciduous broadleaf forest ecosystem in the Mao'ershan watershed. Quantified nitrogen budget and balance across the ecosystem, including atmospheric precipitation, throughfall, stemflow, soil water at different depths, and watershed outflow.

Publications

[1]. Zhang, X.; Liu, J.; Han, H.; Zhang, Y.; Jiang, Z.; Wang, H.; Meng, L.; Li, Y.C.; Liu, Y. Satellite-observed variations and trends in carbon monoxide over Asia and their sensitivities to biomass burning. *Remote Sens.* **2020**, 12, 830, doi:10.3390/rs12050830.

[2]. Zhang, X., J. Liu, Z. Jiang, J. M. Chen, and Y. Zhang (2025), Satellite remote sensing of interannual variation in carbon monoxide over Asia, SPIE, doi: 10.1117/12.3046020.

[3]. Zhang, X.; Liu, J.; Jiang, Z.; Chen H.L.; Tong X.; Chen J.M.; Zhang Y.G. Impact of strong biomass burning on cities in Northeast China: air pollution, health risk, and correlations among pollutants. **202602** (Under review at ***Atmosphere Environment***).

[4]. Zhang, X.; Chen, H.L.; Zhang Y.Z. Quantifying transport-induced uncertainties in global methane inversions: a multi-model evaluation using GEOS-Chem and TM5 constrained by surface and satellite observations. **202606** (To be submitted ***ACP***).

[5]. Zhang, X.; Liu, J. ENSO teleconnection to interannual variability in carbon monoxide in Asia: Investigating the relationship with biomass burning. **202603** (To be submitted ***JGR-A***).

[6]. Li, Y.; Liu, J.; Han, H.; Zhao, T.; Zhang, X.; Zhuang, B.; Wang, T.; Chen, H.; Wu, Y.; Li, M. Collective impacts of biomass burning and synoptic weather on surface $PM_{2.5}$ and CO in Northeast China. *Atmos. Environ.* **2019**, 213, 64–80, doi:10.1016/j.atmosenv.2019.05.062.

[7]. Han, H.; Liu, J.; Yuan, H.; Wang, T.; Zhuang, B.; Zhang, X.; Foreign influences on tropospheric ozone over East Asia through global atmospheric transport. *Atmos. Chem. Phys.* **2019**, 19, 12495-12514, doi:10.5194/acp-19-12495-2019.

[8]. Liu, Y.; Liu, J.; Xie, M.; Fang, K.; Tarasick, D. W.; Wang, H.; Meng, L.; Cheng, X.; Han, H.; Zhang, X. ENSO teleconnection to interannual variability in carbon monoxide over the North Atlantic European Region in spring, *Front. Environ. Sci.* **2020** 10, 894779, doi:10.3389/fenvs.2022.894779.

Academic Presentations

Asia-Pacific Remote Sensing 2024.12 Conference Paper Kaohsiung, Taiwan

Satellite remote sensing of interannual variation in carbon monoxide over Asia

AGU Fall Meeting 2019 Oral Presentation California, USA

Satellite-measured long-term variations of atmospheric carbon monoxide over Asia and their sensitivity to Biomass burning

2019 ISPRS Workshop on Remote Sensing and Synergic Analysis on Atmospheric Environment Poster

Annual Meeting of Atmospheric Environment Branch, Chinese Society for Environmental Sciences 2019 Poster

Technical Skills

Programming: Python, R, Matlab, C, Fortran, NCL

Proficient in Linux system operations and experienced with AWS cloud computing platforms.

Atmospheric Models: TM5, GEOS-Chem, FLEXPART, WRF, HYSPLIT

English Proficiency: CET-6